

Mahmoud Maher El-Said

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PROFESSIONAL SUMMARY

Artificial Intelligence student specializing in Intelligent Systems, with applied portfolio projects in Python, data analysis, machine learning, computer vision, reinforcement learning, and RAG systems. Comfortable working with datasets, notebooks, image-processing pipelines, model evaluation, and clear technical documentation. Seeking remote internships, junior AI/Data roles, and freelance opportunities.

TECHNICAL SKILLS

Programming & Tools: Python, Jupyter Notebook, Git, GitHub, VS Code, HTML/CSS/JavaScript basics

Data Analysis: Pandas, NumPy, data cleaning, EDA, data visualization, Microsoft Excel, Matplotlib

Machine Learning: scikit-learn, supervised learning, unsupervised learning, model evaluation, feature engineering, classification, regression

Deep Learning & Computer Vision: TensorFlow/Keras, PyTorch basics, OpenCV, CNNs, image processing, segmentation, localization

Reinforcement Learning & AI Applications: Q-Learning, SARSA, First-Visit Monte Carlo, Webots, Gymnasium, RAG, embeddings, Chroma, prompt design

SELECTED PROJECTS

Financial Report RAG Assistant | *Python, Chroma, Embeddings, Gemini, RAG*

- Built a retrieval-augmented generation workflow for financial report question answering, including document loading, chunking, embeddings, retrieval, prompt building, and response generation.
- Used Chroma and local embeddings to retrieve relevant document chunks and support source-aware answers.
- Structured tests for core prompt-building and chunking logic to improve project reliability and documentation.

Skin Lesion Segmentation and Localization | *Computer Vision, U-Net, TensorFlow/Keras, OpenCV*

- Developed a medical image analysis workflow for skin lesion segmentation using image and mask preparation, preprocessing, training, validation, and visual outputs.
- Organized predicted masks, before-and-after visualizations, and training artifacts to support clear project documentation.

Sudoku Solver from Camera | *OpenCV, Image Processing, Digit Recognition, Backtracking*

- Created an end-to-end pipeline to detect a Sudoku board from an image, correct perspective, extract cells, recognize digits, solve the puzzle, and visualize the result.
- Applied thresholding, contour detection, perspective transform, grid extraction, and rule-based solving.

Reinforcement Learning GridWorld with Webots | *Q-Learning, SARSA, Monte Carlo, Webots*

- Implemented and compared Q-Learning, SARSA, and First-Visit Monte Carlo in a grid-world environment with obstacles, rewards, episodes, and policy tracking.
- Integrated learned behavior into a Webots simulation and logged actions, rewards, and path progress to explain agent decisions.

SimpleML Library from Scratch | *NumPy, Machine Learning Fundamentals, Reusable Code*

- Built a small educational machine learning library from scratch to practice algorithm implementation, training logic, and reusable component design.

CIFAR-10 Image Classifier API | *CNN/ResNet, FastAPI, Python*

- Prepared an image classification workflow and a simple FastAPI inference interface, practicing preprocessing, evaluation, and deployment-oriented project structure.

EDUCATION

Arab Academy for Science, Technology and Maritime Transport

Egypt

Bachelor of Artificial Intelligence, Intelligent Systems Specialization

Expected Graduation: 2027

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Data Science, Python Programming, Intelligent Systems, Data Analysis.

CERTIFICATIONS AND TRAINING

AMIT Beyond Education – Master Machine Learning & AI Diploma

Jan 2025 – Aug 2025

220 hours; Grade 92%; Python, Data Science, Machine Learning, Deep Learning, Computer Vision, NLP

Coursera / MBRF / UNDP – Machine Learning L1 Badge

Aug

Supervised learning, unsupervised learning, neural networks, model evaluation, TensorFlow, scikit-learn, RL fundamentals

LANGUAGES

Arabic: Native

English: Professional Working Proficiency